Number Sequences (Set B)

Year 4 · **Homework Worksheet**

Worked Example

Example — Mixed Sequence Practice: What comes next? 3, 7, 11, 15, __ and what type of sequence is this (arithmetic or geometric)?

Answer: 19 comes next (gap +4 throughout). This is an arithmetic sequence because the terms increase by a constant amount.

Quick Practice (15 questions)

1. What comes next? 5, 10, 15, 20, __
2. What comes next? 90, 80, 70, 60, __
3. What comes next? 3, 6, 12, 24, __ A) 30 B) 36 C) 42 D) 48 E) 54
4. Select ALL that apply: which of these sequences increase by exactly 3 each time? A) 1, 4, 7, 10 B) 2, 5, 9, 12 C) 0, 3, 6, 9 D) 5, 8, 10, 13 E) 10, 13, 16, 19
5. What comes next? 8, 16, 24, 32, __
6. What comes next? 81, 72, 63, 54, __ A) 44 B) 45 C) 46 D) 47 E) 48
7. What comes next? 1, 8, 27, 64, __ (these are cube numbers)
8. Fill in the missing number: 4, 8, __, 16, 20 (increasing by 4 each time)
9. What comes next? 3, 6, 12, 24, 48, __ A) 72 B) 84 C) 90 D) 96 E) 100
10. Select ALL that apply: which of these are true about the sequence 2, 6, 18, 54? A) Each term is multiplied by 3. B) Each term increases by 4. C) The next term would be 162. D) The sequence is geometric. E) The sequence is arithmetic.
11. What comes next? 4, 9, 14, 19, __
12. What comes next? 2, 2, 4, 6, 10, 16, __ (Fibonacci-style: add the two previous terms) A) 20 B) 22 C) 24 D) 26 E) 28
13. What comes next? 70, 63, 56, 49, __
14. Fill in the missing number: 3, 6, 12, __, 48 (each term doubles)
15. Select ALL that apply: which statements about the sequence 2, 4, 6, 8, 10 are true? A) These are even numbers. B) The next term is 12. C) Each term increases by a constant amount of 2. D) The gap between terms increases each time. E) The sequence is decreasing.

Mixed Practice (15 questions)

1. What comes next? 3, 5, 9, 15, 23, __ (the gap increases by 2 each time: +2, +4, +6, +8...)
2. What comes next? 162, 54, 18, 6, __
3. What comes next? 1, 4, 5, 9, 14, __ (Fibonacci-style: add the two previous terms)
4. A sequence follows the rule "double, then add 2". Starting at 2, what is the 4th term? (2, 6, 14, __)
5. Select ALL that apply: which of these sequences follow a Fibonacci-style rule? A) 2, 3, 5, 8, 13 B) 3, 6, 9, 12, 15 C) 1, 1, 2, 3, 5, 8 D) 4, 4, 8, 12, 20, 32 E) 2, 4, 8, 16, 32

© PassFrame Tuition — for registered student use

6. What comes next? 6, 9, 15, 24, 36, __ (the gap increases by 3 each time: +3, +6, +9, +12...)

7. A sequence follows the two-step rule "multiply by 3, then subtract 2". Starting at 2, what are the first four terms?

8. What comes next in this sequence of triangular numbers? 1, 3, 6, 10, 15, 21, __

9. In the sequence 2, 5, 10, 17, 26, __, what is the rule linking consecutive terms, and what comes next?

10. Select ALL that apply: which of these sequences have gaps that themselves increase by a constant amount? A) 2, 3, 5, 8, 12 (gaps 1,2,3,4) B) 5, 10, 15, 20 (gaps all 5) C) 1, 2, 4, 7, 11 (gaps 1,2,3,4) D) 3, 9, 27, 81 (gaps 6,18,54) E) 4, 7, 11, 16, 22 (gaps 3,4,5,6)

11. What comes next? 405, 135, 45, 15, __

12. A sequence starts 3, 4 and each new term is found by adding the previous two terms. What are the next three terms after 3, 4?

13. What comes next? 1, 3, 12, 60, __ (hint: each term is multiplied by an increasing amount: ×3, ×4, ×5...)

14. Select ALL that apply: which of these are geometric sequences? A) 5, 10, 20, 40 B) 4, 8, 12, 16 C) 200, 100, 50, 25 D) 3, 9, 27, 81 E) 2, 4, 6, 8

15. A sequence follows the rule "add 3, then multiply by 2". Starting at 1, find the next two terms after 1.

Standard / Advanced (15 questions)

1. A sequence follows the rule 'double the number, then add 2'. Starting at 3, what is the next number in the pattern?

2. What number comes two terms after the next term in the sequence: 4, 9, 14, 19, ?

3. A sequence follows the rule 'subtract 2, then double the result'. Starting at 9, what is the next number in the pattern? A) 12 B) 14 C) 16 D) 18 E) 20

4. What number comes two terms after the next term in the sequence: 50, 43, 36, 29, ?

5. A sequence follows the rule 'add 3, then double the result'. Starting at 2, what is the next number in the pattern?

6. A sequence is defined by: term = (previous term × 2) - 1. If the first term is 3, find the first four terms of the sequence.

7. What comes next in this Fibonacci-style sequence (each term is the sum of the two before it)? 2, 3, 5, 8, 13, ? A) 18 B) 19 C) 20 D) 21 E) 22

8. What comes next in this sequence of triangular numbers? 1, 3, 6, 10, 15, ?

9. Select ALL that apply: for the sequence 4, 8, 16, 32, which of these are true? A) Each term is multiplied by 2. B) Each term increases by 4. C) The next term would be 64. D) The sequence is geometric. E) The sequence is arithmetic.

10. A sequence follows the rule "double the number, then subtract 3". Starting at 5, what is the 3rd term? (5, 7, __)

11. What comes next? 3, 5, 8, 12, 17, ? (the gap increases by 1 each time: +2, +3, +4, +5...) A) 21 B) 22 C) 23 D) 24 E) 25

12. A sequence starts 1, 3 and each new term is found by adding the previous two terms. What are the next three terms after 1, 3?

13. What comes next in this sequence, which alternates two simple operations: 6, 10, 8, 12, 10, ? (hint: alternately add 4 and subtract 2)

14. Select ALL that apply: which of these are geometric sequences? A) 3, 6, 12, 24 B) 4, 8, 12, 16 C) 80, 40, 20, 10 D) 2, 5, 8, 11 E) 1, 4, 16, 64

© PassFrame Tuition — for registered student use

15. A sequence follows the rule "add 2, then double the result". Starting at 3, find the next two terms after 3.

© PassFrame Tuition — for registered student use